

[Total Suspended Solids Sensor] [Operation Manual]



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1. Overview

Smart TSS sensor, using RS485 communication interface and standard Modbus protocol, with cleaning brush. Use infrared LED with wavelength of 860nm as light source, unaffected by chroma of water sample, 45° scattering method is adopted, comply with ISO 7027 international standard and USEPA 180.1 environmental standards for the United States. Included with data analysis software, with calibration, recording, analysis, diagnosis and other functions. The sensor has the advantages of self-cleaning, maintenance-free, anti-interference ability.

1.1 Product Features

- With self-cleaning function, remove attachments from water samples
- Digital modulation filter technology, eliminate ambient light effects
- Bubble compensation algorithm, reducing bubble interference in water samples
- More than 10 years long life with infrared LED light source
- Corrosion resistant housing, waterproof grade is IP68, can work underwater for a long time
- RS485 communication interface, standard Modbus protocol, easy to be integrated
- Data analysis software, with calibration, recording, analysis, diagnosis functions.

1.2 Main uses and scope of application

Widely used in surface water, municipal sewage, industrial wastewater, sewage treatment, water plant, process control and other fields.

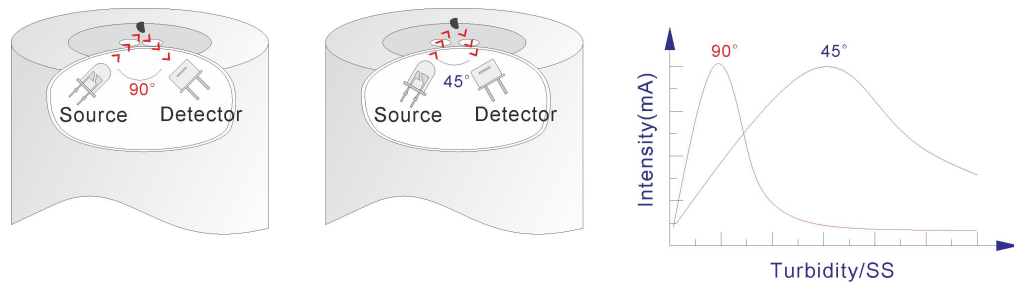
1.3 Environmental conditions for use

Temperature: (-5-40) °C ; pressure: (0-5) bar

2. Working Principle

Suspended solid refers to the solid substance suspended in water. It is one of the indicators to measure the degree of water pollution and the main cause of water

turbidity. As the particle concentration increases, the incident light cannot illuminate all particles, and the scattered light in the 90 ° direction is blocked by other particles and cannot be detected. Therefore, 45 ° scattered light is used to measure the suspended solids.



3. Technical Specifications

3.1. Main performance

Measured parameters	Suspended Solids		
Working principle	Double beam scattering method		
Emission wavelength	860nm		
Test range	(1.0-1000.0) mg/L	(1.0-10.0) g/L	(1.0-25.0) g/L
Resolution	1 mg/L	0.01 g/L	0.01 g/L
Measurement accuracy	±2% FS		
Linearity error	< 5%		
Communication interface	RS485, standard Modbus protocol		
Dimensions	D30mm, L185mm, cable 3 meters (can be customized)		
Working condition	(-5~40°C), (0-5) bar		
Working voltage	12V/24V DC		

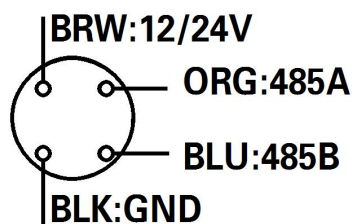
3.2. Dimension



4. Use, Operation

4.1. Wiring instructions

The sensor external interface has 4 lines, correspond to BRW (brown) connect 12V or 24V, BLK (black) connect GND, ORG (orange) connect 485A, BLU (blue) connect 485B. As follows:



4.2. Operation instructions

After connecting to the power, 485 communication and ground lines according to the above, the sensor can be connected by the host computer, etc. The default communication parameter are 9600, 8, 1, N.

4.3. Communication protocol

Item	Register Address	Data Type	Read and Write	Length (Byte)	Remarks
Measured value	0000H	Float	R	4	
Signal value	0100H	Float	R	4	
The first point calibration value	0200H	Float	R	4	
The first point signal value	0202H	Float	R	4	
The second point calibration value	0204H	Float	R	4	
The second point signal value	0206H	Float	R	4	

The third point calibration value	0208H	Float	R	4	
The third point signal value	020AH	Float	R	4	
The first point calibration	1000H	Float	W	4	The first point standard solution value
The second point calibration	1003H	Float	W	4	The second point standard solution value
The third point calibration	1006H	Float	W	4	The third point standard solution value
Slave address	1103H	UINT	R/W	2	1-31
Communication parameters	1104H	UINT	R/W	2	Baud rate : 00:9600 01:19200 02:115200
		UINT	R/W	2	Data bit : 01:8 bit
		UINT	R/W	2	Check bit : 00:None 01:Odd 02:Even
		UINT	R/W	2	Stop bit : 00:1 bit 01:2 bit
Cleaning control	1450H	UINT	W	2	00: start-up
Cleaning cycle	1451H	UINT	R/W	2	0—25hour (0 means 0.5 hour, 25 means Do Not start the brush)
Cleaning mode	1452H	UINT	R/W	2	00:1 turn 01:2 turn 02:4 turn
Factory setting	1500H	UINT	W	2	00: Restore factory setting

5. Instruction Example

1. Read suspended solids value

Send instructions: 01 03 00 00 00 02 C4 0B (01 is communication address, 03 is function code, 0000 is the starting address, 0002 is reading the number of registers, C4 0B is check code)

Return instructions: 01 03 04 E6 66 42 31 DC 10

(01 is communication address, 03 is function code, 04 is the number of bytes returned, E6 66 42 31 is suspended solids 44.47 mg/L , DC 10 is check code)

2. The first point calibration

Send instructions: 01 10 10 00 00 03 06 00 00 00 00 40 A0 83 F8 (01 is communication address, 10 is function code, 1000 is the starting address, 0003 is number of written registers, 06 is number of written bytes, 0000 is reserved bit, 00 00 40 E0 is standard solution value 5.00, 83 F8 is check code)

Return instructions: 01 10 10 00 00 03 84 C8 (01 is communication address, 10 is function code, 1000 is the return register address, 03 is number of registers, 84C8 is check code)

3. The second point calibration

Send instructions: 01 10 10 03 00 03 06 00 00 80 00 45 3B 18 CC (01 is communication address, 10 is function code, 1003 is the starting address, 0003 is the number of written registers, 06 is number of written bytes, 0000 is reserved bit, 80 00 45 3B is standard solution value 3000.00, 18 CC is check code)

Return instructions: 01 10 10 03 00 03 74 C8 (01 is communication address, 10 is function code, 1003 is the return register address, 03 is number of registers, 74C8 is check code)

4. Cleaning control

Send instructions: 01 06 14 50 00 00 8C 2B (01 is communication address, 06 is function code, 1450 is the starting address, 0000 is written value, 8C 2B is check code)

Return instructions: 01 06 14 50 00 00 8C 2B (01 is communication address, 06 is function code, 1450 is the return register address, 0000 is the return written value, 8C 2B is check code)